

As an experienced engineer I bring the following to this role:

- AI fluency in Claude and Copilot and structured approach to agentic programming
- Founder engineer at startup, enterprise experience, high ownership, and ability to move projects from 0 to 1
- Frontend + Backend + Relational databases + DevOps

Experience

Lead Developer | 08/2025 - 12/2025 | **ABC Academy of Music** | MS Industry Capstone Project | Remote (Phoenix, AZ)

Led a team of 5 developers driving agile practices, technical direction, and requirement gathering with the PM.

- Built a low latency frontend using server-side rendering (SSR) of Java Spring Boot REST APIs backed by PostgreSQL.
- Implemented a low-latency notification system using Socket.io, enabling real-time delivery of user alerts with sub-second response times.
- Increased the team's feature delivery velocity by 75% by introducing the team to an agentic workflow that uses detailed JIRA stories as context for Copilot while leading agile practices.

Lead Developer | 04/2023 - 12/2023 | **Tecnotree** | Fulltime | Bengaluru, India

- **Built stepper UI using React. Achieved 20% higher usage than old flow; received "Spotlight Top Performer" award.**
- Developed React components and Node.js APIs consistently and reliably earning accolades.

Senior Software Engineer | 11/2021 - 03/2023 | **PangeaTech** | Fulltime | Bengaluru, India

Founding engineer who built a greenfield ML product from 0 to 1, owning performance, infrastructure, and deployment.

Optimized MySQL database refresh for client Tableau dashboards:

- **Engineered a 97% reduction in MySQL database refresh duration (from 8 hours to 15 minutes)** by converting synchronous Python to asynchronous Python, saving significant client time.
- Reduced failure rates from 50% to <10% by throttling the API call rate to comply with API rate limits and periodically flushing data to database using SQL bulk insert to prevent OOM exceptions.
- Maintained MySQL servers at 99% availability, debugged and maintained SQL queries, triggers, and stored procedures.

Built data visualization dashboards

- Used Highcharts and Chart.js in the React frontend app to create 25+ highly customized charts to enhance data visualization for clients powered by data from MongoDB, served by an Express.js API service (Node.js).
- Reduced API calls by 25% by using Redux and React-Query to cache results.

Scaled data processing of training datasets:

- Reduced data processing time by 80% (4 hours to 45 minutes) by redesigning the Python ML backend to distribute workloads across Kubernetes nodes, saving customer time.
- Reduced Azure cloud costs 20% with better Azure Virtual Machine utilization. Scaled Node.js (Express.js app) across CPU cores using PM2 cluster mode, running one process per core.

Built a distributed logging system:

- Replaced the need for manual server login and exception trace inspection with a searchable, real-time Kibana dashboard, significantly reducing debugging time for pipeline errors (from 4 hours to real time).
- Enhanced code quality and standards by writing 50+ code reviews and mentoring junior developers.

Software Engineer | 07/2019 - 04/2021 | **Tenerity (a J.P. Morgan and Chase company)** | Fulltime | Pune, India

Worked on a high-traffic travel loyalty platform serving millions of customers keeping security and maintainability in mind.

- Increased click through rate by 15% by switching to a new UI for cross-sell itineraries developed with Angular.
- Migrated a .NET C# REST API service to GraphQL reducing network calls by more than 25%.
- Increased test coverage to 90% and ensured compliance with accessibility standards (WCAG) and internationalization.

Projects

Compiler, developing [compiler.app](#), a simulation-based system design learning platform (Leetcode for system design).

- Built a simulation engine with TypeScript using prompt-driven development relying on:
 - i. Detailed class specification written on top of the class file (specifications acts as context)
 - ii. Corresponding unit test which tests for each specification (tests act as guardrails for the AI agent)

iii. AI Agent implements the class using the above as prompt context

- Developing classes according to OOPs and SOLID design principles and using design patterns.
- Built a custom renderer engine in React for animation using custom React Hooks, Context API, SVG and CSS.
- Managed infrastructure and secure hosting (TLS), SEO for visibility, nginx as a reverse proxy, and site reliability.

Music Recommendation and Classification, a RAG and LLM based music discovery system

- Built a GenAI-powered music recommender using PostgreSQL + pgvector for embedding-based semantic retrieval, with LLM-driven song summarization and recommendations.

Travel Recommendations MCP, a Model Context Protocol server to help AI agents better tailor travel recommendations

- Provides JSON-RPC based MCP endpoints to help AI agents provide better travel recommendations across offers.
- Typescript written REST API service which serves tailored recommendations considering location, loyalty points, pagination, client offerings, previous bookings and previous offer clicks.
- Fault-tolerant service with fallbacks on service non-availability, rate limiting to handle excess requests and which points Agentic client to alternate resources on failure.

Python Board Game, a game written in Python with custom graphics and audio effects.

- Received the “Best Paper Award”, a university level award for my paper “Indian Sub-Continent Risk Game”.

Additional Experience

HCI Researcher | 01/2026 - Present | **Arizona State University** | Volunteer | Tempe, AZ

- Developing an Android app that generates context-aware replies using LLMs.

Full-stack Engineer Intern | 09/2021 - 10/2021 | **Friendly Brands** | Internship | Remote

- Improved click-through rate of an SSR rendered upsell UI widget by 20% for an e-commerce live site through data-driven A/B testing of alternative UI features using Python (FastAPI).

Technical Skills

- **Full-stack Technologies:** Node.js, Express.js, TypeScript, Python, Java (Spring), React, Redux, JavaScript, CSS, HTML
- **Database Technologies:** MySQL, PostgreSQL, MongoDB, SQL, Elasticsearch
- **DevOps, Infra & Cloud:** Docker, GitHub Actions, Git, Linux, Azure (VM, Service Bus, Blob Storage), AWS (EC2), nginx

Education

- **Master of Science in Software Engineering**, 3.87 / 4.0 GPA | 12/2025 | **Arizona State University** | Mesa, AZ
- **Bachelor of Science in Computer Science** | **REVA University** | Bengaluru, India